1. Brief Description of the Solution (Max 250 Words):

OnSite360 is an open-source, mobile-first construction project management platform designed to connect field teams and office staff seamlessly. It addresses critical challenges such as fragmented communication, project delays, and the high cost of proprietary software. The platform provides a cost-effective and fully customizable alternative that suits organizations of all sizes.

Targeted at field workers, engineers, project managers, supervisors, executives, and clients, OnSite360 automates project workflows, digitizes documentation, and delivers real-time insights to speed up decision-making and reduce costly delays.

A standout feature is the AI Copilot integration, allowing organizations to connect GPT or Gemini APIs. This intelligent assistant helps users—especially engineers, site supervisors, and project managers—ask questions about documents, generate project summaries, and interactively analyze data.

Built with React for the frontend and NestJS for the backend, OnSite360 supports scalability. It uses PostgreSQL as the database, and with Docker, it's easy to deploy on private servers. PWA support ensures offline access, crucial for construction sites with limited connectivity.

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2. Functionality and Features (Max 350 Words):

OnSite360 offers a range of functionalities designed to streamline construction project workflows while catering to specific user roles:

- **Real-Time Communication:** Integrated chat and notification system.
- **Document Management:** Centralized platform to upload, approve, and access project documents.
- **Task Automation:** Auto-assignment, approval workflows, and deadline reminders.
- **AI Copilot:** GPT/Gemini-powered assistant for document queries, data analysis, and summarization.
- **Offline Mode:** PWA enables uninterrupted access in low-network areas.
- **Custom Dashboards:** Role-specific insights for executives, supervisors, and engineers.
- **Role-Based Permissions:** Fine-grained access control for every user role.

Targeted Users & User Requirements: System Admins, Executive Admins, Project Directors, Engineers, Site Supervisors, Subcontractors, and Clients. The system allows each role to have a personalized interface and functionality relevant to their responsibilities.

Scalability & Expandability: Modular architecture allows easy scaling and integration with third-party APIs. Companies can deploy the system on cloud or on-premise infrastructure using Docker.

User-Friendliness & Compatibility: The system uses a modern UI built with React and is optimized for both desktop and mobile users. It integrates well with tools like Jira, GitHub, and calendar services.

Security Features: Includes JWT-based authentication, encrypted data storage, and audit logs. Access is controlled via strict RBAC models.

Maintenance-Related Features: Built-in health checks, modular code for easy updates, and detailed logs for issue tracking.

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3. Product Description – Proof of Concept (Max 350 Words):

OnSite360 has successfully undergone internal pilot testing with simulated projects involving real-world scenarios. These trials included project documentation uploads, real-time collaboration, and testing offline support at a mock construction site environment.

Feedback from early users—primarily undergraduate engineering students, instructors, and industry mentors—showed notable improvements in documentation accuracy and team coordination. With AI Copilot, users quickly found relevant data and summarized multi-page reports, saving hours of manual reading.

The product's open-source nature and Docker support made it easy to deploy on local university servers and test in isolated environments. Its modular design allowed quick customization for various test projects, proving its adaptability.

OnSite360 demonstrated commercial potential due to increasing demand for affordable, customizable solutions in the construction industry. Most existing platforms (e.g., Procore, Autodesk Build) are costly and not suitable for small to mid-sized firms, especially in developing regions. OnSite360 fills this gap.

Future pilots are planned in collaboration with small local construction firms and academic research centers, focusing on real-world deployments and further refining AI use cases.

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4. Product Description – Uniqueness (Max 350 Words):

OnSite360 stands out due to its open-source model, AI integration, and offline-first architecture. Unlike many international solutions like Procore or Buildertrend, which are costly and closed-source, OnSite360 is free to use and customizable for any organization.

Unique Technologies & Features: - **AI Copilot:** GPT/Gemini-powered assistant embedded directly into the platform. - **Fine-Grained RBAC:** Customizable permissions matching real organizational hierarchies. - **Offline PWA Support:** Full mobile functionality in low-connectivity zones. - **Modular Architecture:** Easy plugin-style extension system.

Problems Solved: - Eliminates disconnected site-to-office workflows. - Reduces high costs of commercial PM tools. - Bridges communication gaps with role-specific interfaces.

Similar Products: - **Procore (US)** – Enterprise-focused, expensive. - **Autodesk Build (Global)** – Powerful, but lacks offline support and flexibility. - **Fieldwire (Asia Pacific)** – Limited customization, closed-source.

Why OnSite360 is Unique in the Region: - First open-source tool with both offline access and AI integration for South Asian construction. - Tailored for developing economies and mid-sized firms.

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5. Product Description – Quality and Application of Technology (Max 350 Words):

Technology Stack: - Frontend: React + Vite - Backend: NestJS (Node.js) - Database: PostgreSQL - DevOps: Docker - PWA Support: Service Workers

Quality Assurance Activities: - Manual and automated testing (Jest + Cypress) - Internal UAT with real-world use cases - Static code analysis with ESLint and Prettier

Standards Followed: - Secure coding guidelines (OWASP) - Docker container security practices

Product Stability & Feedback: Test users reported smooth real-time collaboration, stable offline behavior, and intuitive role-based UIs. No major bugs occurred during multi-day use tests.

Packaging & Deployment: - Docker image with `docker-compose.yml` for one-click local or cloud deployment. - Documentation site (Markdown & Docusaurus) - REST API docs via Swagger UI

Awards: - (N/A for now – under pilot/testing phase)

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6. Product Description – Previous Participation (Max 350 Words):

This version of OnSite360 has not participated in any APICTA competition previously.

However, an early prototype was presented as a university group project and was recognized internally for its innovation and practical value.

Since then, the platform has undergone significant updates: - **AI Integration Added:** GPT/Gemini Copilot - **Offline PWA Support Implemented** - **Role-Based Dashboards and RBAC System Introduced** - **Improved UI/UX with Tailwind + Figma Redesign**

The current version is a complete rewrite, with scalable backend architecture, modern frontend stack, and full documentation. It is now mature enough for commercial deployment and national/international competitions.